BSCS THESIS TOOL REVISION

THESIS TITLE	SECTION	GROUP NO.
BOPIS: Bayesian Optimization and Pareto-Based Intelligent Configuration Selection for Energy-Efficient Local LLM Inference	BSCS 4-5	2

Thesis/Tool Part	COMMENTS/SUGGESTIONS/RECOMMENDATIONS by the Panel	ACTION TAKEN by the Proponent/s	Page No.	REMARKS (Panelist Approval by indicating the date)			
				Panel 1	Panel 2	Panel 3	Panel 4
Title*	NO COMMENT						
Preliminaries	NO COMMENT						
Chapter 1*	- Clearly indicate the strength, innovation, and novelty of BOPIS - Indicate in the Statement of the Problem that the inputs are not predefined, to avoid showing bias.	- Justified random search as the baseline: it is a practical, reproducible standard for comparing optimization methods (Bergstra & Bengio, 2012) (p. 10). - Stated the priority of the system: BOPIS selects the configuration with the greatest energy reduction, and speed and quality act as minimum thresholds (SRR, QRR), not competing goals (p. 11). - Showed where BOPIS is better by comparing per metric (energy, speed, quality, resource use) and per precision variant (F16, Q8_0, Q4_K_M) with the Wilcoxon signed-rank test (Statement of the Problem, pp. 14–15). - Defined success criteria (EIR, SRR, QRR) so “better than the baseline” is a measurable result, and stated how null, negative or partial outcomes are reported (Scope and Limitation, p. 17).					

		- Pinpointed the weaknesses of BOPIS in Scope and Limitation (pp. 16–18): ● Energy is estimated in software, which adds measurement variance. ● GPU offloading slowed generation on this GPU class. ● Device discovery covers NVIDIA devices only. ● Results apply to one model, one machine and one dataset. ● Extending device discovery is listed as future work.					
Chapter 2*	- Justify the use of random search as the baseline - Highlight the novelty of BOPIS in comparison with existing studies	- Justified random search as an uninformed, impartial baseline that needs no prior knowledge of the configuration space (p. 47). - Stated the novelty of BOPIS: no reviewed study combines Bayesian Optimization and Pareto analysis in one offline, pre-deployment system covering runtime parameters and GGUF precision variants for local inference (Synthesis, p. 51). - Explained the cost: BOPIS runs once before deployment and its overhead does not affect response time in production (p. 47).					
Chapter 3*	- Identify the specific parameters in which BOPIS performs better than random search, and the strengths that supports this						

	d more datasets to test whether BOPIS performs better - State why BOPIS is more expensive than other models						
Tool (Software/Hardware)	- Explain how BOPIS optimizes energy use. - Show an actual use case where a real user inputs the question. - Show the priority of the system (e.g. energy consumption, resource utilization, etc.)						
References	NO COMMENT						
General Observations	- The proposed model has to perform better (even on a single parameter) than the baseline. - Pinpoint the weaknesses of BOPIS and improve from there.						

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Name of Panelist 3: Mariel Leo T. Violeta **Name of Panelist 4:**

***Revisions should a way that it will not change the initially approved proposal and that proposal changes should NOT be accommodated in order to satisfy the inadequacies of the tool created.**